Distinguishing Technical Problem Solving and General Discontent in Reddit Software Update Discussions

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***Abstract*—Reddit hosts large-scale software update discussions where troubleshooting requests and general frustration often appear together. This makes automatic interpretation difficult. We present a framework that separates Technical Problem Solving (TPS) from General Discontent (GDS) and jointly analyzes author-community sentiment divergence in discussion threads. This distinction is important because software update discussions often contain both actionable troubleshooting reports and general complaints, making it difficult for developer teams to quickly identify discussions that require technical attention. We construct a corrected TPSGDS-542 corpus ($n = 542$; 86 TPS, 456 GDS) and evaluate rule-based, classical machine learning, and transformer models using stratified five-fold cross-validation. Multinomial Naive Bayes provides the best minority-class tradeoff, with Macro-F1 $= 0.659$ and TPS-F1 $= 0.463$. We further find that TPS threads show significantly higher author-community sentiment divergence than GDS threads (0.371 vs. 0.287, $p = 0.009$). These results suggest that TPS filtering can help developer teams more efficiently identify troubleshooting discussions within large volumes of update-related feedback, while sentiment divergence provides additional insight into how communities respond to technical issues.**

***Index Terms*—Reddit, discourse classification, sentiment divergence, software update discussions, Technical Problem Solving, General Discontent, social media mining**

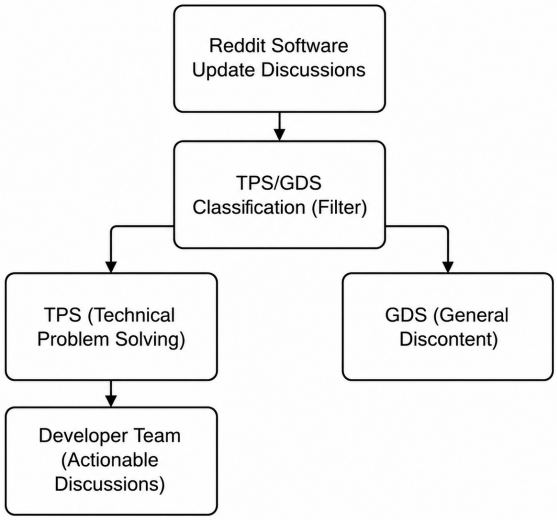


Fig. 1. Conceptual overview of TPS filtering for software update discussions. The framework separates troubleshooting-oriented discussions from general complaints, helping developer teams focus on actionable technical information.

I. Introduction

Social media has become one of the main places where people discuss products, services, and technical problems. In 2025, global social media users reached 5.24 billion, which is about 63.9% of the world population [1]. Reddit is one of the largest discussion-based social platforms, with millions of active users and thousands of topic-focused communities in 2025 [2]. Because users post quickly and interact in threads, Reddit provides rich real-world text data for studying how people react to software changes. Software update discussions are especially valuable because they include direct user experiences after releases. Users report bugs, installation failures, feature confusion, and performance drops, while others share workarounds and practical fixes. At the same time, many posts express anger, fatigue, or disappointment without clear technical details. This difference matters. A post that asks for a fix should not be treated the same as a post that mainly expresses frustration. In this paper, we call these two discourse types *Technical Problem Solving* (TPS) and *General Discontent* (GDS). By automatically identifying TPS discussions, developer teams can quickly locate troubleshooting reports and actionable feedback within large volumes of software update discussions, reducing the effort required to manually sift through general complaints.

These discussion patterns create challenges for automated analysis systems that attempt to interpret user reactions consistently. Sentiment analysis has progressed from lexicon-driven methods to machine learning and deep learning approaches [3], [4]. Reddit has also been widely used for mining public opinion, topic patterns, and health-related or education-related discussions [5]–[8]. In addition, conversation-level studies have shown that interaction structure can reveal meaningful stance dynamics in online threads [9]. Technical support research has also shown that text classification can improve issue handling and response workflows [10].

However, existing studies largely treat actionable software feedback and general user dissatisfaction as part of broader classification tasks while rarely modeling author–

Code and dataset: https://github.com/SE4CPS/Research-on-Software-Updates/tree/open-text-sentimental-sync/research-open-text-sentimental

community interaction throughout discussion threads. These limitations motivate a framework that jointly distinguishes troubleshooting-oriented discourse from general discontent while modeling author–community interaction over time [11].

To address these gaps, we propose a TPS/GDS framework for Reddit software update discussions that combines discourse classification with author–community sentiment trajectory analysis. The main contributions are:

- Introduce TPSGDS-542, a manually corrected Reddit benchmark for distinguishing Technical Problem Solving (TPS) from General Discontent (GDS).
- Compare rule-based, classical machine learning, and transformer-based approaches for TPS/GDS classification.
- Propose an author–community sentiment trajectory framework to quantify emotional divergence throughout discussion threads.
- Evaluate robustness through feature ablations, software-focused subreddit analysis, and independent human validation.

II. Related Work

Reddit has become an important source for studying large-scale online discussions because of its topic-focused communities, threaded conversations, and publicly available user interactions [5]. Prior studies have applied Reddit data to opinion mining, topic discovery, software support, complaint analysis, issue categorization, software requirements engineering, and sentiment analysis, demonstrating its value for understanding user-generated discussions across domains [3], [4], [6], [7], [10], [12]–[14]. However, existing studies primarily focus on software feedback categorization or general sentiment analysis rather than distinguishing troubleshooting-oriented discussions from general dissatisfaction or modeling author–community interaction.

Conversation structure and temporal interactions provide information beyond isolated text analysis. Reply chains capture stance and evidential relationships, while longitudinal sentiment analysis reveals how emotions evolve throughout online discussions [9], [11]. However, existing studies rarely distinguish troubleshooting-oriented software discussions from general user dissatisfaction or jointly analyze discourse type with author–community emotional interaction. Our work addresses this gap through integrated TPS/GDS classification and author–community sentiment divergence analysis.

III. Methodology

Figure 2 presents the overall experimental workflow, which consists of three stages: (1) construction of a manually validated Reddit corpus with corrected gold labels, (2) TPS/GDS discourse classification using rule-based and supervised learning methods, and (3) author–community sentiment trajectory analysis for measuring discussion-level emotional divergence.

A. Reddit Corpus and Gold Labels

Software-update-related Reddit discussions were collected using a open source software update monitoring platform that provides access to Reddit posts through its API. Candidate threads were retrieved from TPS-oriented and GDS-oriented filtered pools. These platform-generated categories were used only for candidate retrieval and not as ground-truth labels. Similar Reddit-based data collection strategies have been adopted in prior social media mining studies [7].

Each retrieved thread was manually reviewed and assigned one of two discourse labels: Technical Problem Solving (TPS) or General Discontent (GDS). The resulting TPSGDS-542 corpus contains 542 Reddit threads, comprising 86 TPS and 456 GDS discussions. Approximately 24% of the original platform-assigned labels differed from the manually corrected labels, demonstrating the limitations of heuristic filtering for identifying troubleshooting-oriented discussions.

For each thread, we retained the post title, discussion text, and metadata required for classification and sentiment trajectory analysis.

B. Text Construction for Classification

For supervised classification, we constructed multiple text representations from each Reddit thread. As shown in Fig. 2, the primary representation, `text_raw`, combines the post title, body, and up to ten highest-scored Reddit comments to preserve discussion context [7], [15].

For feature ablations, we generated title-only and title+body variants. Classical machine learning models used the cleaned text representation, whereas VADER and RoBERTa operated on the original discussion text [4], [16].

C. TPS/GDS Classification Models

We evaluated four discourse classification approaches using stratified five-fold cross-validation. A majority-class baseline that always predicts GDS was included to quantify the effect of class imbalance. Logistic Regression and Multinomial Naive Bayes were trained using TF-IDF features with unigrams, bigrams, sublinear term frequency scaling, and class-balanced weights, following common practice in text classification [17].

RoBERTa-base was fine-tuned on `text_raw` using a maximum sequence length of 256, batch size of 8, learning rate of 2×10^{-5}, and two training epochs per fold [16]. For comparison, a rule-based VADER baseline was applied to `text_raw` using a `technical_first` policy with troubleshooting keywords such as *error*, *crash*, and *fix* [4]. Discussions with strongly negative sentiment but no technical indicators were mapped to GDS, using a neutral compound-score threshold of ± 0.15.

D. Author–Community Trajectory Construction (C1′)

To analyze interaction dynamics beyond discourse classification, we constructed C1′ sentiment trajectories from the complete chronological Reddit comment tree contained in the raw API JSON data rather than the reduced TF-IDF representation.

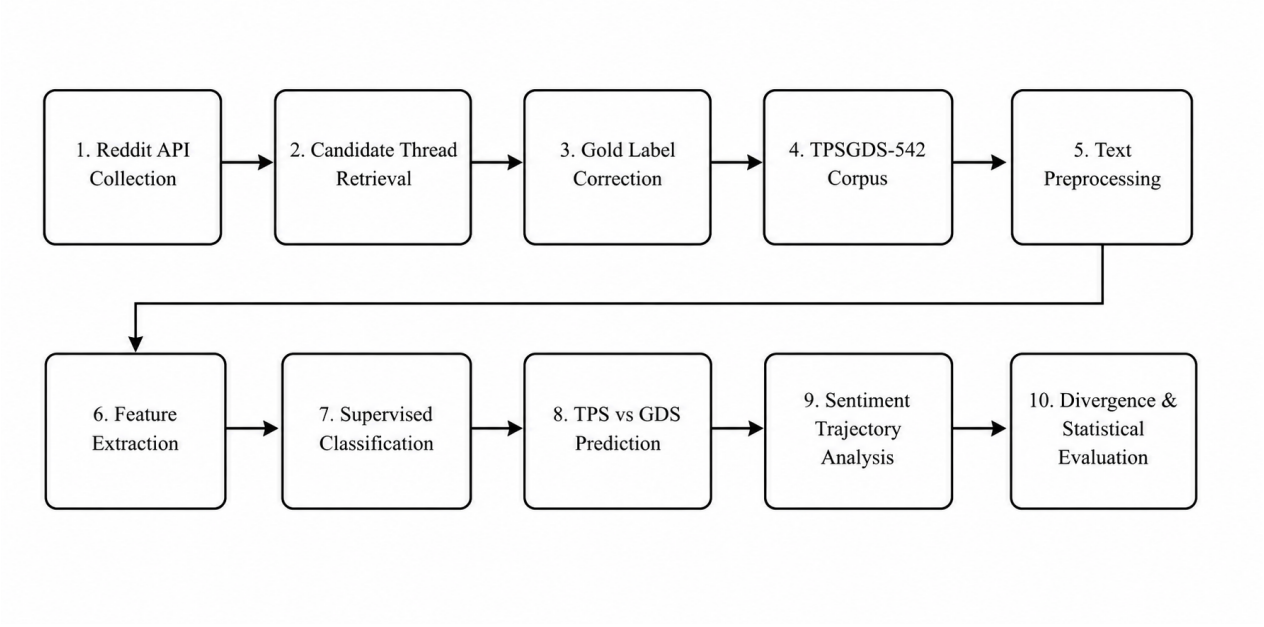


Fig. 2. Overview of the proposed TPS/GDS analysis pipeline. Reddit software-update discussions are collected through a commercial monitoring platform, manually validated to obtain corrected gold labels, transformed into textual representations, and analyzed using discourse classification and author–community sentiment trajectory methods.

Comments were ordered according to the `created_utc` timestamp and assigned VADER compound sentiment scores [4]. Author-side utterances consisted of the original post body together with all replies written by the original submitter, whereas community-side utterances included all remaining non-empty comments.

For both author and community trajectories, we extracted summary statistics including mean sentiment, sentiment trend, step-wise volatility, and early-versus-late sentiment shift for divergence analysis and feature ablation experiments. When no author reply was available, the sentiment score of the post title was used as a fallback.

E. Sentiment Divergence and Evaluation

Using the C1′ author–community sentiment trajectories, we quantified sentiment divergence as the absolute difference between the mean author and community VADER compound sentiment scores [4]. Differences between TPS and GDS discussions were evaluated using Welch's t-test, the Mann–Whitney U test, Cohen's d, and bootstrap confidence intervals. Robustness analyses were additionally performed on a software-development subreddit subset. VADER scores were interpreted as relative indicators of author–community emotional divergence rather than direct measures of affect [15].

Classification performance was evaluated using stratified five-fold cross-validation on the 542-thread corpus with out-of-fold metrics including accuracy, macro-F1, TPS-F1, precision, recall, and PR-AUC [18]. Feature ablations compared title-only, title + body, full-thread TF-IDF, trajectory-enhanced, and subreddit-aware representations. Label reliability was assessed on 80 stratified Reddit threads using Cohen's κ agreement statistic [19].

TABLE I
CLASSIFICATION PERFORMANCE ON 542 LABELED REDDIT THREADS (5-FOLD STRATIFIED CV, OOF).

Model	Acc.	Macro-F1	TPS-F1	TPS Prec.	TPS Rec.	PR-AUC
Majority	0.841	0.457	0.000	0.000	0.000	0.159
Naive Bayes	0.771	0.659±0.035	0.463±0.047	0.372	0.616	0.411
Logistic Reg.	0.845	0.632±0.087	0.352±0.162	0.490	0.278	0.431
RoBERTa-base	0.793	0.591±0.084	0.307±0.189	0.279	0.359	0.391
VADER rules	0.421	0.409±0.042	0.330±0.027	0.202	0.896	0.191

IV. RESULTS

A. TPS/GDS Classification Benchmark

The rule-based VADER baseline performs poorly overall despite high TPS recall (0.896). Its low TPS precision (0.202) indicates that many negative or technical-sounding posts are incorrectly classified as troubleshooting discussions. This pattern is consistent with the label-correction step, where heuristic API labels were often noisy.

Among the learned models, Multinomial Naive Bayes achieves the strongest balance between overall performance and minority-class recovery, obtaining the highest Macro-F1 (0.659) and TPS-F1 (0.463). Logistic Regression achieves the highest overall accuracy (0.845) but under-recovers TPS discussions, with TPS recall dropping to 0.278. RoBERTa-base does not outperform Naive Bayes in this setting, likely because of the relatively small TPS class and the domain-specific characteristics of Reddit language.

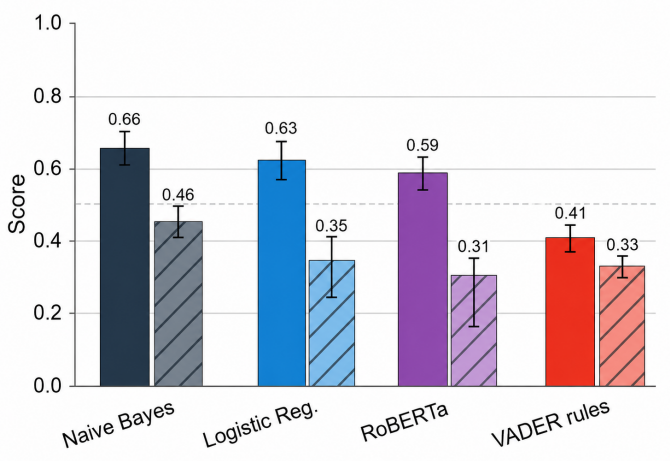


Fig. 3. OOF macro-F1 and TPS-F1 for four classifiers on the full corpus (n=542). Error bars represent fold standard deviation.

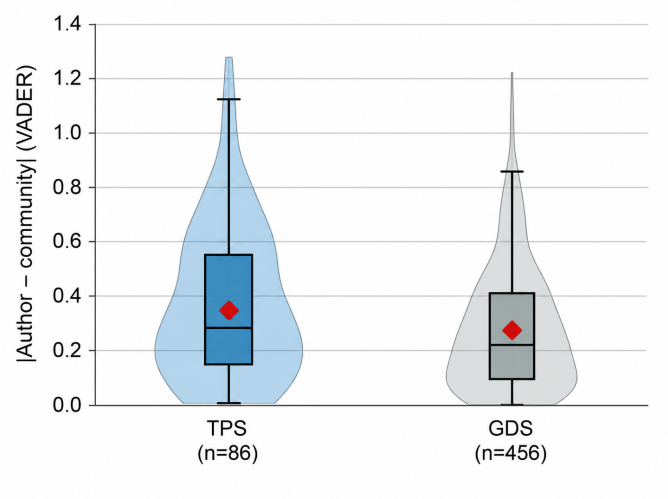


Fig. 4. Distribution of thread-level sentiment divergence (VADER) for TPS (n=86) and GDS (n=456).

TABLE II
C1′ DIVERGENCE STATISTICS: TPS VS. GDS (FULL CORPUS).

Measure	TPS	GDS	Test
Mean divergence	0.371	0.287	MW p=0.009
Author more negative (%)	69.8	53.7	p=0.006
Cohen's d		0.33	
95% CI		[0.023, 0.146]	

The Naive Bayes confusion profile indicates a recall-oriented classifier that identifies most TPS discussions while accepting additional false positives.

B. Author Community Sentiment Divergence

TPS discussions exhibit significantly greater author–community sentiment divergence than GDS discussions (Table II; Figs. 4 and 5).

Original posters in TPS discussions were more often negative than the surrounding community.

Figure 5 shows a consistently larger author–community sentiment gap for TPS discussions.

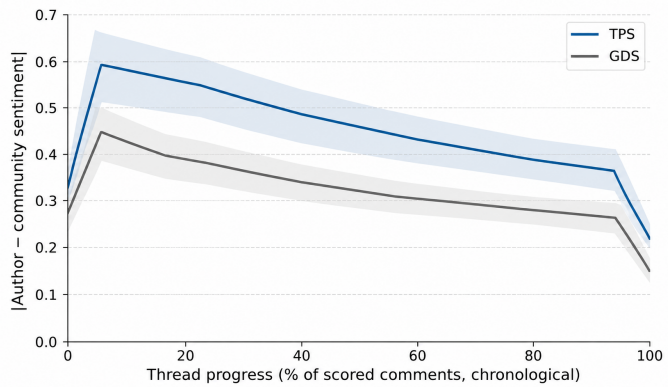


Fig. 5. Mean running author community gap vs. normalized thread progress (shaded: ±1.96 SEM across threads).

C. Feature Ablations, Robustness, and Label Validation

We evaluate how different input representations affect TPS/GDS classification (Table III).

The title-only representation (D1) achieves the best overall performance, outperforming the full-thread TF-IDF representation (D3) in both Macro-F1 and TPS-F1. This suggests that Reddit post titles capture the strongest discriminative signal for distinguishing troubleshooting-oriented discussions from general discontent. In contrast, later comments often introduce unrelated conversations, humor, or repeated advice that dilute the discriminative power of the TF-IDF representation.

Adding trajectory features (D4) improves TPS-F1 (0.352→0.423), showing that author-community interactions complement textual features. Including subreddit metadata (D5) yields only marginal improvement, suggesting that model performance is not primarily driven by subreddit-specific memorization.

Table IV shows that the primary findings remain stable on Tier-A software and developer subreddits, with improved Naive Bayes performance and a slightly larger sentiment divergence gap, supporting the robustness of the framework.

To assess annotation reliability, an independent annotator reviewed 80 stratified Reddit threads. Agreement with the corrected gold labels reached 91.2%, with Cohen's κ=0.688, indicating substantial agreement. Most disagreements occurred in borderline cases, such as update announcements or ambiguous bug-report discussions.

Overall, the experimental results support three main findings. First, TPS/GDS discourse classification can be learned effectively using classical machine learning models. Second, TPS discussions exhibit significantly stronger author–community sentiment divergence than GDS discussions. Third, Reddit post titles provide the strongest discriminative signal for classification, while trajectory-based features and Tier-A robustness analysis further strengthen the overall conclusions.

TABLE III
FEATURE ABLATIONS (LOGISTIC REGRESSION, 5-FOLD CV, OOF).

ID	Setting	Macro-F1	TPS-F1
D1	Title only	**0.758**±0.060	**0.580**±0.101
D2	Title + post body	0.646±0.037	0.381±0.067
D3	Full thread (TF-IDF)	0.632±0.087	0.352±0.162
D4	Full thread + trajectory	0.647±0.099	0.423±0.162
D5	Full thread + subreddit	0.633±0.070	0.400±0.111

TABLE IV
TIER-A SOFTWARE-SUBREDDIT ROBUSTNESS (5-FOLD CV UNLESS NOTED).

Check	Full (n=542)	Tier-A (n=304)
NB macro-F1	0.659	0.703
NB TPS-F1	0.463	0.535
Divergence Δ (TPS−GDS)	0.084	0.105
Divergence test (p)	0.009	0.012
D1 macro-F1	0.758	0.695
D3 macro-F1	0.632	0.622

V. CONCLUSION

This paper presented a labeled benchmark and empirical study of Technical Problem Solving (TPS) and General Discontent (GDS) in software-update Reddit discussions using a corrected corpus of 542 threads (86 TPS and 456 GDS). Learned models consistently outperformed the rule-based VADER baseline. Among all evaluated approaches, Multinomial Naive Bayes with TF-IDF achieved the best minority-class tradeoff, obtaining a Macro-F1 of 0.659, TPS-F1 of 0.463, and TPS recall of 0.62. Fine-tuned RoBERTa-base did not surpass this classical baseline at the current dataset scale.

Beyond discourse classification, we analyzed author–community sentiment divergence using C1′ VADER trajectories and found that TPS discussions exhibit significantly greater divergence than GDS discussions (0.371 vs. 0.287; Mann–Whitney $p = 0.009$). Original posters in TPS discussions were more frequently more negative than the surrounding community (69.8% vs. 53.7%), and this separation remained consistently higher throughout most stages of thread progression.

Feature ablation experiments further showed that Reddit post titles provide the strongest discriminative signal for TPS/GDS classification, while trajectory-based features improve minority-class detection beyond full-thread text alone. Tier-A software-subreddit robustness experiments and independent human validation (91.2% agreement; $\kappa = 0.688$) further support the reliability of the proposed framework.

Overall, the results demonstrate that troubleshooting-oriented and discontent-oriented software-update discussions can be distinguished effectively using discourse-aware NLP while exhibiting distinct author–community interaction dynamics. Future work will expand the framework to additional platforms and evaluate its practical usefulness with software engineering teams.

TABLE V
HUMAN VS. GOLD TPS/GDS LABELS (n=80).

	Human TPS	Human GDS
Gold TPS	10	3
Gold GDS	4	63

Accuracy 91.2%; κ=0.688.

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